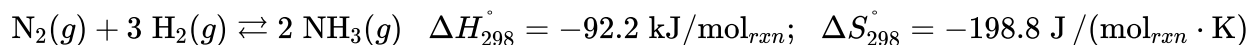


Chapter 10 Part 2: Le Chatelier's Principle FRQ

1. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to NH_3 .



(a) The reaction of $\text{N}_2(g)$ and $\text{H}_2(g)$ to form $\text{NH}_3(g)$ is represented above. The reaction has been studied in order to maximize the yield of $\text{NH}_3(g)$.

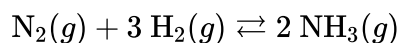
- Calculate the value of ΔG° , in kJ/mol_{rxn} , at 298 K.
- Calculate the value of the equilibrium constant, K_p , at 298 K.

Bond	Average Bond Enthalpy (kJ/mol)
N – H	391
H – H	436
N $\equiv$ N	?

(b) Based on the average bond enthalpies in the table above and the value of ΔH_{298}° , calculate the value of the N $\equiv$ N bond enthalpy.

(c) Although the reaction is thermodynamically favorable at 298 K, it does not occur to an observable extent at this temperature. Explain how the N $\equiv$ N bond enthalpy affects the rate of the reaction.

(d) The reaction is usually carried out at a very high temperature in the presence of a catalyst. Explain why the reaction is thermodynamically unfavorable at very high temperatures. Justify your answer in terms of ΔH° , ΔS° , and their effect on ΔG° .



(e) Although the forward reaction represented above is thermodynamically unfavorable at very high temperatures, it can be used to produce NH_3 if the NH_3 is continually removed from the reaction vessel. Explain in terms of Q and K_p why this method works.

The boiling points of H_2 and NH_3 are given in the table below.

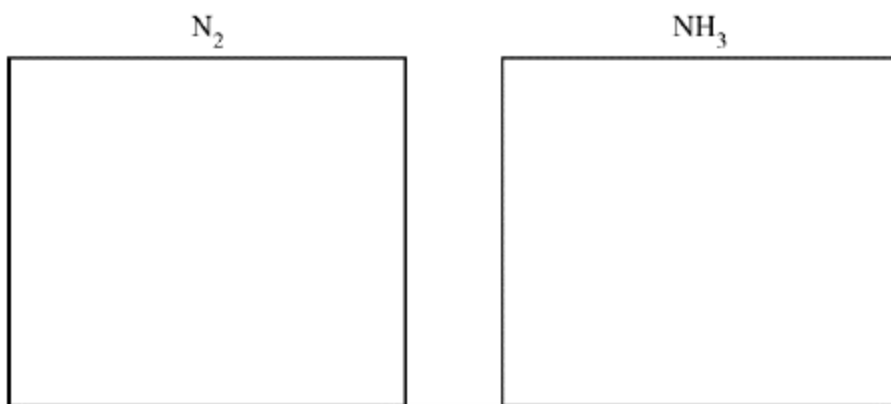
Substance	Boiling Point
H_2	20 K
NH_3	240 K

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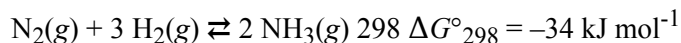
(f) A student claims that $\text{H}_2(g)$ condenses at a much lower temperature than $\text{NH}_3(g)$ because H_2 has stronger intermolecular forces than NH_3 does. There are two errors in the student's reasoning. Identify the student's errors and explain your answer in terms of the specific types of forces involved.

2. Answer the following questions about nitrogen, hydrogen, and ammonia.

a. In the boxes below, draw the complete Lewis electron-dot diagrams for N_2 and NH_3 .



b. Calculate the standard free-energy change, ΔG° , that occurs when 12.0 g of $\text{H}_2(g)$ reacts with excess $\text{N}_2(g)$ at 298 K according to the reaction represented below.



c. Given that ΔH°_{298} for the reaction is $-92.2 \text{ kJ mol}^{-1}$, which is larger, the total bond dissociation energy of the reactants or the total bond dissociation energy of the products? Explain.

d. The value of the standard entropy change, ΔS°_{298} , for the reaction is $-199 \text{ J mol}^{-1}\text{K}^{-1}$. Explain why the value of ΔS°_{298} is negative.

e. Assume that ΔH° and ΔS° for the reaction are independent of temperature.

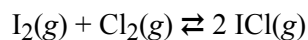
- i. Explain why there is a temperature above 298 K at which the algebraic sign of the value of ΔG° changes.
- ii. Theoretically, the best yields of ammonia should be achieved at low temperatures and high pressures.

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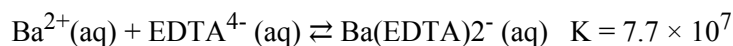
3. Answer the following questions about the thermodynamics of the reactions represented below.



- Is reaction X, represented above, spontaneous under standard conditions? Justify your answer with a calculation.
- Calculate the value of the equilibrium constant, K_{eq} , for reaction X at 25°C.
- What effect will an increase in temperature have on the equilibrium constant for reaction X? Explain your answer.
- Explain why the standard entropy change is greater for reaction Y than for reaction X.
- Above what temperature will the value of the equilibrium constant for reaction Y be greater than 1.0? Justify your answer with calculations.
- For the vaporization of solid iodine, $\text{I}_2(s) \rightarrow \text{I}_2(g)$, the value of ΔH°_{298} is 62 kJ mol^{-1} . Using this information, calculate the value of ΔH°_{298} for the reaction represented below.



4.

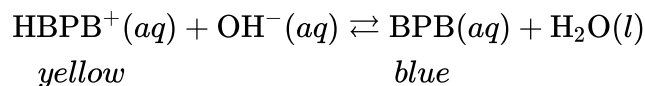


The polyatomic ion $\text{C}_{10}\text{H}_{12}\text{N}_2\text{O}_8^{4-}$ is commonly abbreviated as EDTA^{4-} . The ion can form complexes with metal ions in aqueous solutions. A complex of EDTA^{4-} with Ba^{2+} ion forms according to the equation above. A 50.0 mL volume of a solution that has an $\text{EDTA}^{4-}(\text{aq})$ concentration of 0.30 M is mixed with 50.0 mL of 0.20 M $\text{Ba}(\text{NO}_3)_2$ to produce 100.0 mL of solution.

- Considering the value of K for the reaction, determine the concentration of $\text{Ba}(\text{EDTA})^{2-}(\text{aq})$ in the 100.0 mL of solution. Justify your answer.
- The solution is diluted with distilled water to a total volume of 1.00 L. After equilibrium has been reestablished, is the number of moles $\text{Ba}^{2+}(\text{aq})$ of present in the solution greater than, less than, or equal to the number of moles of $\text{Ba}^{2+}(\text{aq})$ present in the original solution before it was diluted? Justify your answer.

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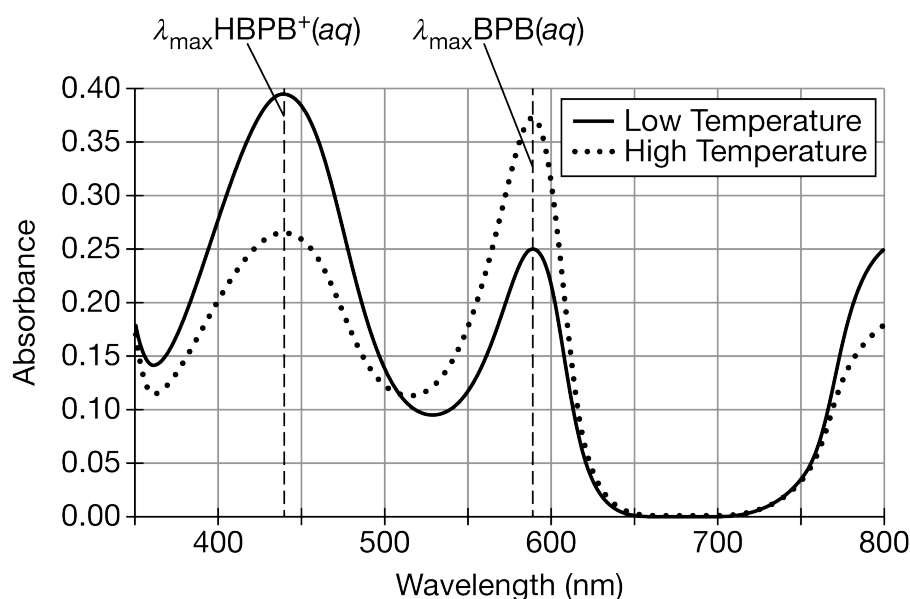
5. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



HBPB^+ , the acidified form of bromophenol blue, BPB, is yellow and has a strong absorption peak at a wavelength of 440 nm. HBPB^+ reacts with hydroxide ions according to the equation above, producing water and BPB, which is blue and has a strong absorption peak at a wavelength of 590 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

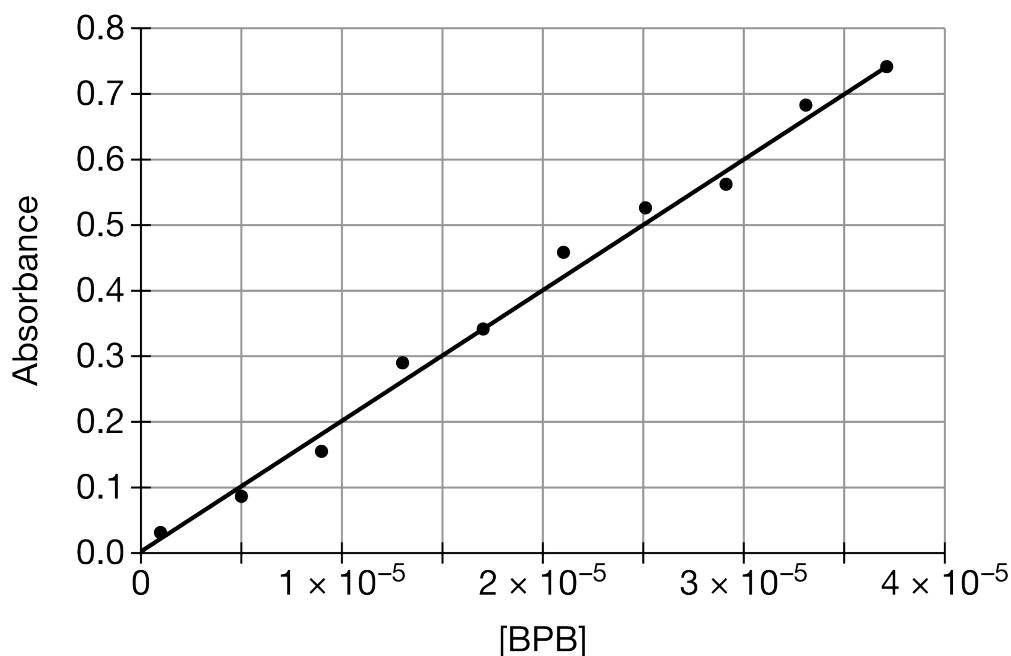
The absorption spectra of an equilibrium mixture of $\text{HBPB}^+(aq)$ and $\text{BPB}(aq)$ collected at two different temperatures, are shown in the diagram below.



- (b) At which wavelength, 350 nm or 440 nm, are photons with the higher energy absorbed? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 590 nm versus the concentration of BPB is shown in the graph below

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(d) Based on the best-fit line shown in the graph, what is the approximate absorbance for a solution in which $[\text{BPB}] = 2.0 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a solution of BPB of unknown concentration in order to determine the $[\text{BPB}]$ in the solution using the calibration plot above. If water remains in the cuvette prior to being filled with the solution, how will the estimated molarity compare to the actual $[\text{BPB}]$ in the solution? Explain your reasoning.

Over time, BPB converts to a colorless molecule in the presence of OH^- ions. To study the rate of the reaction, the student recorded the absorbance of the BPB solution at 590 nm in a 0.10 M NaOH solution. The temperature and $[\text{OH}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

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Time (hr)	Absorbance
0	0.80
0.50	0.56
1.00	0.40
1.50	0.28
2.00	0.20
2.50	0.15
3.00	0.10

- (f) Determine the half-life of the reaction. Include units with your answer.
- (g) Explain how the absorbance and time data are consistent with a process that is first order with respect to BPB.

A second trial is conducted at the same constant temperature, but with $[\text{OH}^-]$ changed to 0.40 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OH}^-]$ was 0.10 M .

- (h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OH}^-]$ is changed from 0.10 M to 0.40 M .
- (i) When $[\text{OH}^-]$ is changed from 0.10 M to 0.40 M , does the magnitude of $\Delta H^\circ_{\text{rxn}}$, the enthalpy change of the reaction, increase, decrease, or stay the same? Justify your answer.

In order to prepare the 0.100 M solution of $\text{NaOH}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 40.0 g of $\text{NaOH}(s)$.

Step 2: Add the $\text{NaOH}(s)$ to a 1000.0 mL volumetric flask that is half-filled with distilled water.

Step 3: Add enough distilled water to bring the volume up to the line on the flask.

- (j) The chemical quantities used in the procedure do NOT produce $0.100\text{ M NaOH}(aq)$. Describe specifically how they should be changed in order to produce $0.100\text{ M NaOH}(aq)$.
- (k) Another student is preparing a 0.100 M NaOH solution using the correct chemical quantities, but notices that some solid remains at the bottom of the volumetric flask. Describe an action that the student should take to ensure that the remaining solid dissolves.
- (l) When making the solution above, the student notices that the temperature increases when dissolving $\text{NaOH}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{NaOH}(s)$ in $1000.\text{ g}$ each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y?

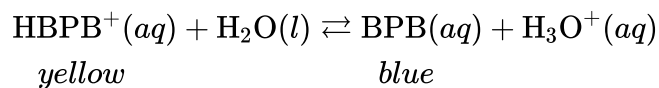
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Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below.

Solvent	X	Y
Specific Heat Capacity (J/(g · °C))	2.53	2.91

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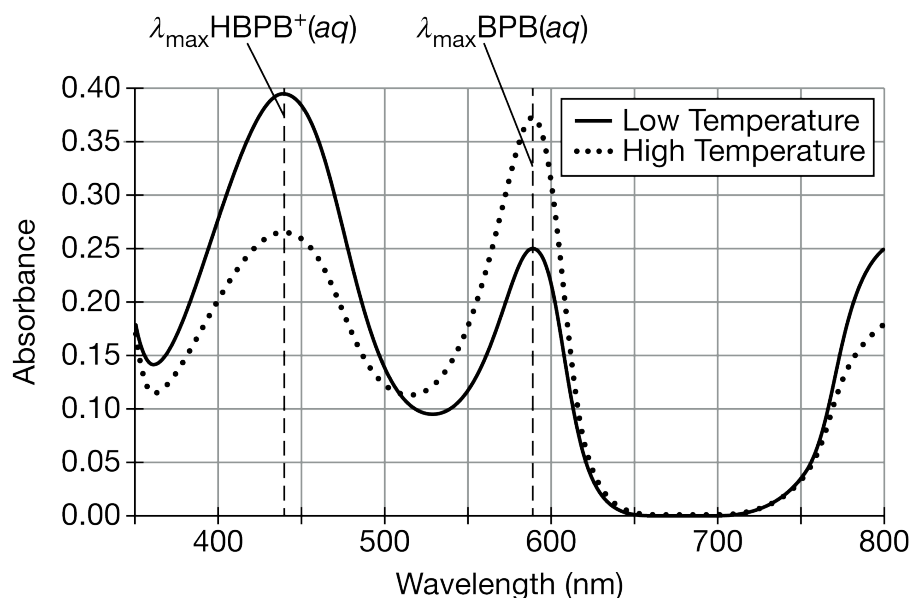
6. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



HBPB^+ , the acidified form of bromophenol blue, BPB, is yellow and has a strong absorption peak at a wavelength of 440 nm. HBPB^+ reacts with water according to the equation above, producing H_3O^+ ions and BPB, which is blue and has a strong absorption peak at a wavelength of 590 nm.

(a) Write the K_{eq} expression for the equilibrium reaction above.

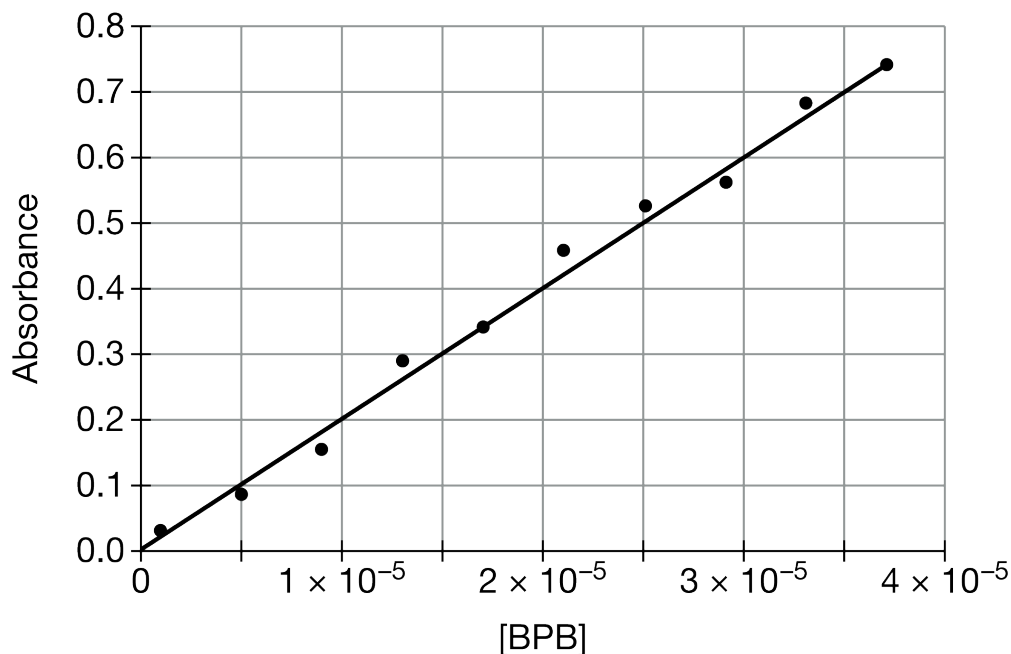
The absorption spectra of an equilibrium mixture of $\text{HBPB}^+(aq)$ and $\text{BPB}(aq)$, collected at two different temperatures, are shown the following diagram.



- (b) Which peak, the one at 590 nm or the one at 800 nm, results from the absorption of photons with the lower energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 590 nm versus the concentration of BPB is shown in the graph below.

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(d) Based on the best-fit line shown in the graph, what is the approximate absorbance of a solution in which $[\text{BPB}] = 1.5 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a solution of BPB of unknown concentration in order to determine the $[\text{BPB}]$ in the solution using the calibration plot above, which was recorded at 590 nm. If the spectrophotometer is set to a wavelength different from 590 nm during the measurement, what would be the effect on the estimated molarity of BPB? Explain your reasoning.

Over time, BPB converts to a colorless molecule in the presence of OH^- ions. To study the rate of the reaction, the student recorded the absorbance of the BPB solution at 590 nm in a 1.0 M NaOH solution. The temperature and $[\text{OH}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

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Time (s)	Absorbance
0	0.60
120	0.42
240	0.30
360	0.21
480	0.15
600	0.11
720	0.08

(f) Determine the half-life of the reaction. Include units with your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to BPB.

A second trial is conducted at the same constant temperature, but with $[\text{OH}^-]$ changed to 0.50 M . The observed half-life of the reaction is longer than for the first trial when $[\text{OH}^-]$ was 1.0 M .

(h) Explain, in terms of particle collisions, why the half-life of the reaction increases when $[\text{OH}^-]$ is changed from 1.0 M to 0.50 M .

(i) Is the activation energy for the reaction when $[\text{OH}^-]$ is 0.50 M greater than, the same as, or less than the activation energy when $[\text{OH}^-]$ is 1.0 M ? Justify your answer.

In order to prepare the 0.500 M solution of $\text{NaOH}(aq)$, the student follows a procedure with the following steps.

Step 1: Measure out 20.0 g of $\text{NaOH}(s)$.

Step 2: Add the $\text{NaOH}(s)$ to a clean, dry 1000.0 mL volumetric flask.

Step 3: Add about 500 mL of distilled water to the flask, and swirl to dissolve the $\text{NaOH}(s)$.

Step 4: Add enough distilled water to bring the mixture to the line on the volumetric flask, inverting several times to mix.

(j) Will the chemical quantities in the procedure produce a solution that is $0.500\text{ M NaOH}(aq)$? Explain your reasoning.

(k) Before adding the $\text{NaOH}(s)$ to the volumetric flask, the student first rinsed the volumetric flask with distilled water, leaving some droplets of distilled water on the inside of the flask. As a result of this step, will the concentration of $\text{NaOH}(aq)$ be greater than, less than, or equal to 0.500 M ? Explain your answer.

(l) When making the solution above, the student notices that the temperature increases when dissolving

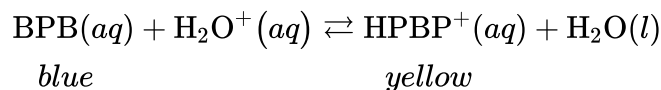
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$\text{NaOH}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{NaOH}(s)$ in 1000. g each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below.

Solvent	X	Y
Specific Heat Capacity (J/(g · °C))	2.91	2.53

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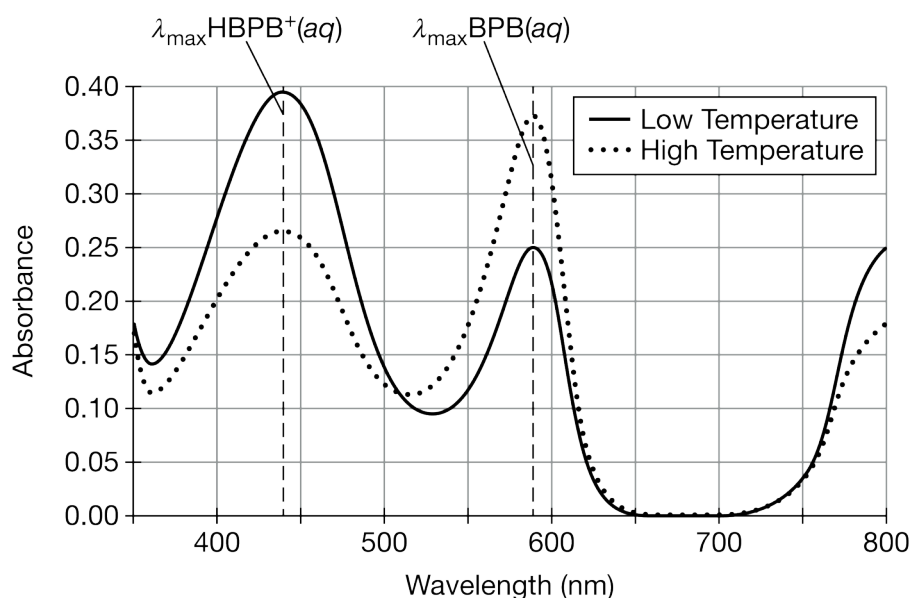
7. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Bromophenol blue, BPB, is a blue dye whose spectrum has a strong absorption peak at a wavelength of 590 nm. BPB reacts with hydronium ions according to the equation above, producing water and HPBP^+ , which is yellow and has a strong absorption peak at a wavelength of 440 nm.

- (a) Write the K_{eq} expression for the equilibrium reaction above.

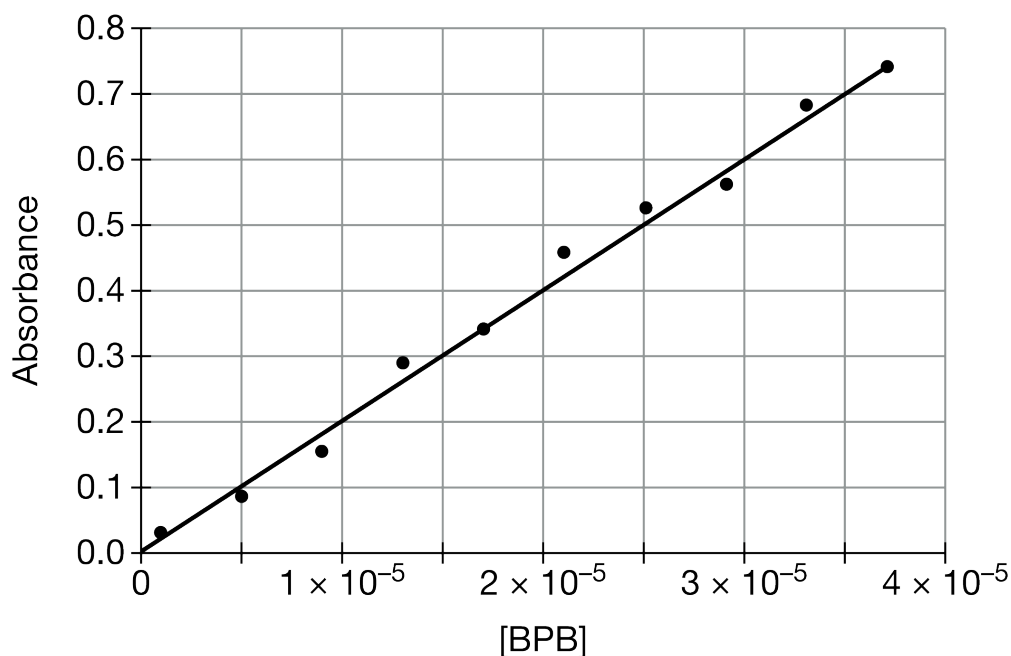
The absorption spectra of an equilibrium mixture of $\text{BPB}(aq)$ and $\text{HPBP}^+(aq)$, collected at two different temperatures, are shown in the following diagram.



- (b) Which peak, the one at 440 nm or the one at 590 nm, results from the absorption of photons with the greater energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 590 nm versus the concentration of BPB is shown in the graph below.

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(d) Based on the best-fit line shown in the graph, what is the approximate absorbance of a solution in which $[\text{BPB}] = 3.5 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a BPB solution of unknown concentration in order to determine the $[\text{BPB}]$ in the solution using the calibration plot above. If a fingerprint remains on the cuvette and in the path of light in the spectrophotometer during the measurement, what would be the effect on the estimated molarity of BPB? Explain your reasoning.

Over time, BPB converts to a colorless molecule in the presence of OH^- ions. To study the rate of the reaction, the student recorded the absorbance of the BPB solution at 590 nm in a 0.50 M NaOH solution. The temperature and $[\text{OH}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

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Time (min)	Absorbance
0	0.60
4	0.48
8	0.38
12	0.30
16	0.24
20	0.19
24	0.15

(f) Determine the half-life of the reaction. Include units with your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to BPB.

A second trial is conducted at the same constant temperature, but with $[\text{OH}^-]$ changed to 2.00 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OH}^-]$ was 0.50 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OH}^-]$ is changed from 0.50 M to 2.00 M .

(i) Is the activation energy for the reaction when $[\text{OH}^-]$ is 2.00 M greater than, the same as, or less than the activation energy when $[\text{OH}^-]$ is 0.50 M ? Justify your answer.

In order to prepare the 2.00 M solution of $\text{NaOH}(aq)$, one student follows a procedure with the following steps.

Step 1: Measure out 16.0 g of $\text{NaOH}(s)$.

Step 2: Add the $\text{NaOH}(s)$ to a 100.0 mL volumetric flask that is half-filled with distilled water.

Step 3: Add enough distilled water to bring the volume up to the line on the flask.

(j) The chemical quantities used in the procedure do NOT produce $2.00\text{ M NaOH}(aq)$. Describe specifically how they should be changed in order to produce $2.00\text{ M NaOH}(aq)$.

(k) Another student is preparing a 2.00 M NaOH solution using the correct chemical quantities, but notices that some undissolved solid remains at the bottom of the volumetric flask. Is the concentration of $\text{NaOH}(aq)$ in the solution in the flask greater than, less than, or equal to 2.00 M ? Explain your answer.

(l) When preparing solutions of NaOH , the student notices that the temperature increases when dissolving $\text{NaOH}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{NaOH}(s)$ in $1000.\text{ g}$ each of two different solvents, X and water. Assuming the magnitude of the enthalpy change is the same regardless of solvent, would the temperature change in X be greater than, less than, or equal to the temperature change in the

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water? Justify your answer. The specific heat capacities of water and solvent X are given in the table below.

Solvent	Water	X
Specific Heat Capacity ($\text{J}/(\text{g} \cdot ^\circ\text{C})$)	4.18	2.53

8. The compound butane, C_4H_{10} , occurs in two isomeric forms, *n*-butane and isobutane (2-methyl propane). Both compounds exist as gases at 25°C and 1.0 atm.
- Draw the structural formula of each of the isomers (include all atoms). Clearly label each structure.
 - On the basis of molecular structure, identify the isomer that has the higher boiling point. Justify your answer.

The two isomers exist in equilibrium as represented by the equation below.



Suppose that a 0.010 mol sample of pure *n*-butane is placed in an evacuated 1.0 L rigid container at 25°C .

- Write the expression for the equilibrium constant, K_c , for the reaction.
- Calculate the initial pressure in the container when the *n*-butane is first introduced (before the reaction starts).
- The *n*-butane reacts until equilibrium has been established at 25°C .
 - Calculate the total pressure in the container at equilibrium. Justify your answer.
 - Calculate the molar concentration of each species at equilibrium.
 - If the volume of the system is reduced to half of its original volume, what will be the new concentration of *n*-butane after equilibrium has been reestablished at 25°C ? Justify your answer.

Suppose that in another experiment a 0.010 mol sample of pure isobutane is placed in an evacuated 1.0 L rigid container and allowed to come to equilibrium at 25°C .

- Calculate the molar concentration of each species after equilibrium has been established.

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9. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about $\text{C}_2\text{H}_4\text{O}$.

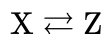
- (a) $\text{C}_2\text{H}_4\text{O}$ (molar mass 44.06g/mol) is a gas at room temperature and can be harmful at concentrations above $8.17 \times 10^{-6} M$. What is the maximum mass of this compound that can safely be present in a room with a volume of $3.00 \times 10^5 \text{L}$?

Two structures, X and Z, with the molecular formula $\text{C}_2\text{H}_4\text{O}$ are shown in the table.

Structure	Lewis Dot Diagram
X	
Z	

- (b) Based on VSEPR theory, predict the $\text{H} - \text{C} - \text{H}$ bond angle in structure X.
- (c) A student incorrectly claims that X and Z are resonance structures of each other. Explain why the student's claim is incorrect.

X and Z are both gases at room temperature and 1atm , and they react according to the following equation.



- (d) Write the expression for the equilibrium constant, K_p , in terms of the partial pressures of X and Z.
- (e) For the reaction represented above, the value of the equilibrium constant, K_p , is 3.3×10^{-8} at 25°C . Which would

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have a greater partial pressure at equilibrium: the reactant, **X**, or the product, **Z**? Circle one of the options and justify your choice.

Option 1 :

The partial pressure of reactant **X** is greater.

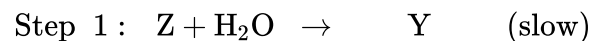
Option 2 :

The partial pressure of product **Z** is greater.

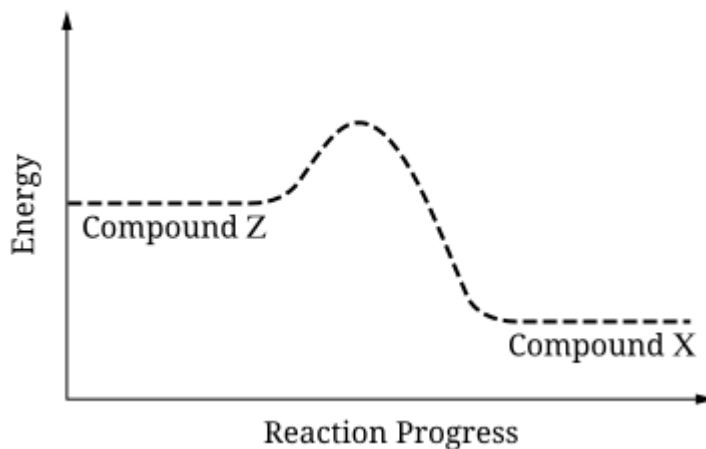
(f) An empty vessel is filled with compound **X** and sealed shut. The initial pressure of compound **X** is 0.55 atm. Calculate the partial pressure of compound **Z** in the vessel after equilibrium is established at constant temperature.

(g) The sample is compressed to half of the original volume, without any change in temperature. Does the value of K_p increase, decrease, or remain the same? Justify your choice.

(h) In the reverse reaction, the conversion of compound **Z** to compound **X** is catalyzed in the presence of trace amounts of water. The catalyzed reaction is proposed to occur via the following two-step mechanism.



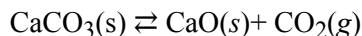
(i) Identify the intermediate in the mechanism, if any.



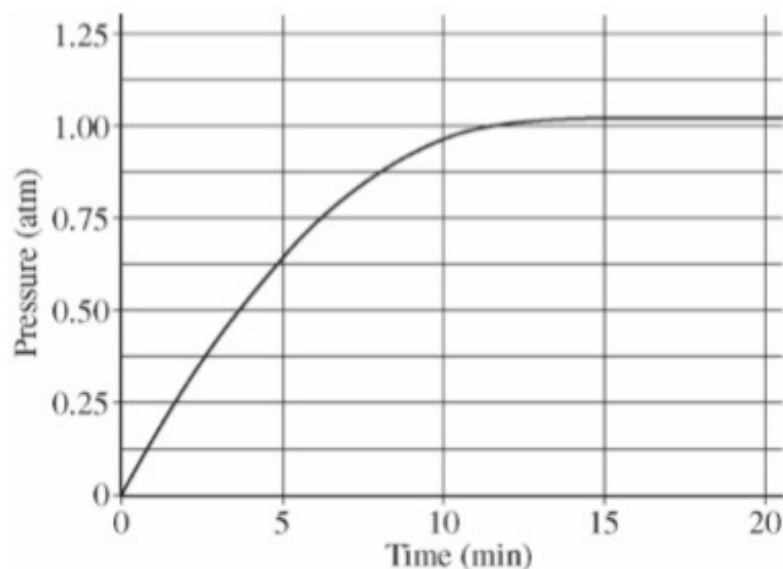
(ii) A reaction profile of the uncatalyzed reaction is shown above. On the same axes, sketch a reaction profile consistent with the proposed mechanism for the catalyzed reaction.

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10.



When heated, calcium carbonate decomposes according to the equation above. In a study of the decomposition of calcium carbonate, a student added a 50.0 g sample of powdered $\text{CaCO}_3(s)$ to a 1.00 L rigid container. The student sealed the container, pumped out all the gases, then heated the container in an oven at 1100 K. As the container was heated, the total pressure of the $\text{CO}_2(g)$ in the container was measured over time. The data are plotted in the graph below.



The student repeated the experiment, but this time the student used a 100.0 g sample of powdered $\text{CaCO}_3(s)$. In this experiment, the final pressure in the container was 1.04 atm, which was the same final pressure as in the first experiment.

- Calculate the number of moles of $\text{CO}_2(g)$ present in the container after 20 minutes of heating.
- The student claimed that the final pressure in the container in each experiment became constant because all of the $\text{CaCO}_3(s)$ had decomposed. Based on the data in the experiments, do you agree with this claim? Explain.
- After 20 minutes some $\text{CO}_2(g)$ was injected into the container, initially raising the pressure to 1.5 atm. Would the final pressure inside the container be less than, greater than, or equal to 1.04 atm? Explain your reasoning.
- Are there sufficient data obtained in the experiments to determine the value of the equilibrium constant, K_p , for the decomposition of $\text{CaCO}_3(s)$ at 1100 K? Justify your answer.

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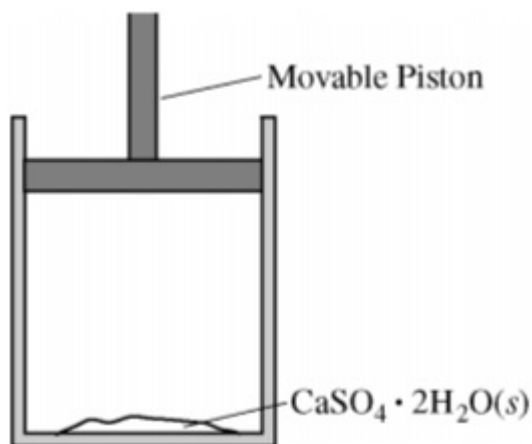
The hydrate $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ can be heated to form the anhydrous salt, $\text{CaSO}_4(s)$, as shown by the reaction represented above.

- a. Using the data in the table below, calculate the value of ΔG° , in $\text{kJ/mol}_{\text{rxn}}$, for the reaction at 298 K.

Substance	ΔG_f° at 298 K (kJ/mol)
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$	-1795.70
$\text{CaSO}_4(s)$	-1320.30
$\text{H}_2\text{O}(g)$	-228.59

- b. Given that the value of ΔH° for the reaction at 298 K is $+105 \text{ kJ/mol}_{\text{rxn}}$, calculate the value of ΔS° for the reaction at 298 K. Include units with your answer.

A sample of $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ is placed in a cylinder with a movable piston as shown in the diagram below. The air above the solid is at 1.00 atm and is initially dry (partial pressure of $\text{H}_2\text{O}(g) = 0 \text{ atm}$).



- c. Write the expression for the equilibrium constant, K_p , for the reaction.
- d. Given that the equilibrium constant, K_p , is 6.4×10^{-4} at 298 K, determine the partial pressure, in atm, of water vapor in the cylinder at equilibrium at 298 K.
- e. If the volume of the system is reduced to one-half of its original volume and the system is allowed to reestablish equilibrium at 298 K, what will be the pressure, in atm, of the water vapor at the new volume? Justify your answer.

In the laboratory, the hydrate $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ can be heated in a crucible to completely drive off the water of hydration to form the anhydrous salt, $\text{CaSO}_4(s)$.

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- f. A 2.49 g sample of pure $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ is heated several times until the mass is constant. Calculate the mass, in grams, of the solid that remains after the dehydration reaction is complete.

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12. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{Cl}_2(g)$ and $\text{CO}(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.



- (a) If 7.0 g of $\text{CO}(g)$ is consumed in the reaction with excess $\text{Cl}_2(g)$, how many moles of $\text{COCl}_2(g)$ are produced?
- (b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

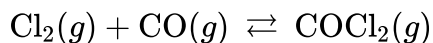
Time	P_{Cl_2} (atm)	P_{CO} (atm)	P_{COCl_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.78	0.78	0.22
t_2	0.62	0.62	0.38
t_3	0.51	0.51	0.49
t_4	0.45	0.45	0.55
t_5	0.42	0.42	0.58
t_6	0.41	0.41	0.59
t_7	0.40	0.40	0.60
t_8	0.40	0.40	0.60
t_9	0.40	0.40	0.60

- (c) At time t_8 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.
- (d) At equilibrium, the container holds the most molecules of which gas, $\text{Cl}_2(g)$ or $\text{COCl}_2(g)$? Explain your answer.

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(e) A student hypothesizes that if the temperature of the container is decreased after time t_9 , the mole fraction of Cl_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.



(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{Cl}_2(g)$ or $\text{COCl}_2(g)$, will deviate most from the ideal gas law at low temperature? Justify your choice.

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13. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{Cl}_2(g)$ and $\text{O}_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.



- (a) If 6.4 g of $\text{O}_2(g)$ is consumed in the reaction with excess $\text{Cl}_2(g)$, how many moles of $\text{Cl}_2\text{O}(g)$ are produced?
- (b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

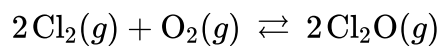
The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{Cl_2} (atm)	P_{O_2} (atm)	$P_{\text{Cl}_2\text{O}}$ (atm)
t_0	0.50	0.50	0.0
t_1	0.42	0.46	0.08
t_2	0.36	0.43	0.14
t_3	0.32	0.41	0.19
t_4	0.29	0.39	0.22
t_5	0.27	0.38	0.23
t_6	0.26	0.37	0.24
t_7	0.25	0.38	0.25
t_8	0.25	0.38	0.25
t_9	0.25	0.38	0.25

- (c) At time t_4 , is the rate of the reverse reaction greater than, less than, or equal to the rate of the forward reaction? Justify your choice.
- (d) At equilibrium, the container holds fewer molecules of which gas, $\text{Cl}_2(g)$ or $\text{O}_2(g)$? Explain your answer.
- (e) A student hypothesizes that if the temperature of the container is decreased after time t_9 , the mole fraction of $\text{Cl}_2\text{O}(g)$ in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.
- (f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by

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the equation below.



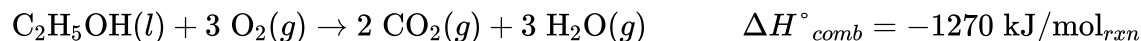
(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{Cl}_2(g)$ or $\text{O}_2(g)$, will deviate most from the ideal gas law at low temperature? Justify your choice.

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14. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Ethanol, $\text{C}_2\text{H}_5\text{OH}$, will combust in air according to the equation above.

- (a) Is $\text{O}_2(g)$ oxidized in the reaction, or is it reduced? Justify your answer in terms of oxidation numbers.
- (b) When a sample of $\text{C}_2\text{H}_5\text{OH}$ was combusted, the volume of $\text{CO}_2(g)$ produced was 18.0 L when measured at 21.7°C and 1.03 atm. Determine the number of moles of $\text{CO}_2(g)$ that was produced.
- (c) Determine the volume of $\text{C}_2\text{H}_5\text{OH}(l)$, in mL, that was combusted to produce the volume of $\text{CO}_2(g)$ collected in part (b). (The density of $\text{C}_2\text{H}_5\text{OH}(l)$ is 0.79 g/mL.)
- (d) Determine the amount of heat, in kJ, that is released by the combustion reaction.

The combustion reaction occurred in a closed room containing 5.56×10^4 g of air originally at 21.7°C . Assume that all of the heat produced by the reaction was absorbed by the air (specific heat = $1.005 \text{ J/(g}^\circ\text{C)}$) in the room.

- (e) Determine the final temperature of the air in the room after the combustion.
- (f) The boiling points of ethanol and dimethyl ether are provided in the following table. Identify the intermolecular force that is most responsible for the difference in boiling point. In terms of this force, explain why the boiling point of ethanol is higher than that of dimethyl ether.

Compound	Structural Formula	Boiling Point ($^\circ\text{C}$)
Ethanol	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	78
Dimethyl ether	$\begin{array}{c} \text{H} \quad \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \quad \text{H} \end{array}$	-24

One method of producing ethanol is by the reaction of ethene and water, as represented below.



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(g) A chemist wants to run the reaction and maximize the amount of $\text{C}_2\text{H}_5\text{OH}(g)$ produced. Identify two ways the chemist could change the reaction conditions (other than adding or removing any chemical species) to favor the formation of more product. Justify your answer.

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15. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{F}_2(g)$ and $\text{O}_2(g)$ are placed in a previously evacuated rigid container, where they react according to the equation below.

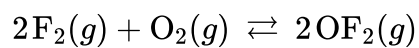


- (a) If 3.8 g of $\text{F}_2(g)$ is consumed in the reaction, how many moles of $\text{O}_2(g)$ will be needed?
- (b) Which element is reduced in this reaction? Justify your answer in terms of oxidation numbers.

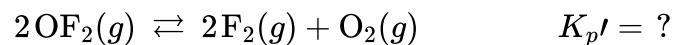
The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{F_2} (atm)	P_{O_2} (atm)	P_{OF_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.79	0.90	0.21
t_2	0.62	0.81	0.38
t_3	0.50	0.75	0.50
t_4	0.41	0.71	0.59
t_5	0.37	0.69	0.63
t_6	0.35	0.68	0.65
t_7	0.34	0.67	0.66
t_8	0.33	0.67	0.67
t_9	0.33	0.67	0.67

- (c) At time t_2 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.
- (d) At equilibrium, the container holds fewer molecules of which gas, $\text{F}_2(g)$ or $\text{O}_2(g)$? Explain your answer.
- (e) A student hypothesizes that if the temperature of the container is increased after time t_9 , the mole fraction of OF_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.
- (f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.

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(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{F}_2(g)$ or $\text{OF}_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

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16. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{H}_2(g)$ and $\text{Br}_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.

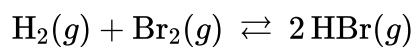


- (a) If 1.6 g of Br_2 is consumed in the reaction with excess H_2 , how many moles of HBr are produced?
- (b) Which element is reduced in this reaction? Justify your answer in terms of the oxidation numbers.

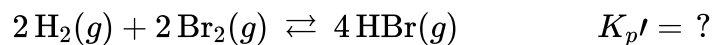
The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{H_2} (atm)	P_{Br_2} (atm)	P_{HBr} (atm)
t_0	1.0	1.0	0.0
t_1	0.78	0.78	0.44
t_2	0.62	0.62	0.76
t_3	0.51	0.51	0.98
t_4	0.45	0.45	1.10
t_5	0.42	0.42	1.17
t_6	0.41	0.41	1.19
t_7	0.40	0.40	1.20
t_8	0.40	0.40	1.20
t_9	0.40	0.40	1.20

- (c) At time t_9 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.
- (d) At equilibrium, the container holds the most molecules of which gas, H_2 , Br_2 , or HBr ? Explain your answer.
- (e) A student hypothesizes that if the temperature of the container is decreased after time t_9 , the mole fraction of HBr in the container will decrease. Do you agree or disagree with the student's hypothesis? Justify your answer.
- (f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.

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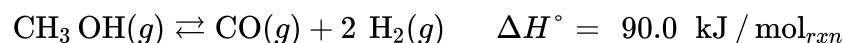
(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{Br}_2(g)$ or $\text{H}_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

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17. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Methanol vapor decomposes to form carbon monoxide gas and hydrogen gas at high temperatures in the presence of a platinum catalyst, as represented by the balanced chemical equation given.

(a) Are the hydrogen atoms oxidized or are they reduced in the forward reaction? Justify your answer in terms of oxidation numbers.

(b) In the following box, draw the complete Lewis electron-dot diagram for the carbon monoxide molecule in which every atom obeys the octet rule. Show all bonding and nonbonding valence electrons.



(c) The values of the standard molar entropies of the compounds involved in the reaction are given in the following table.

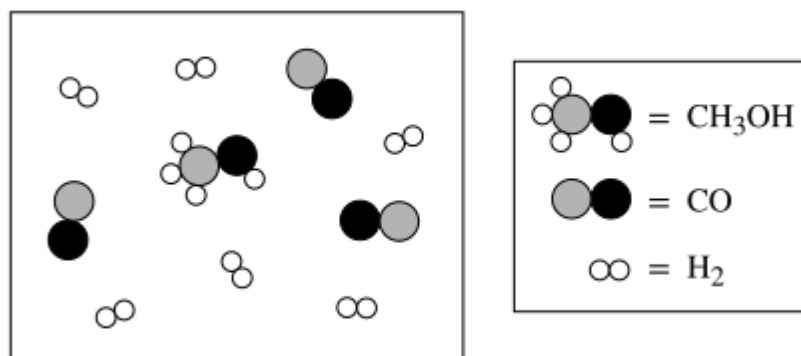
Substance	S° (J/(K·mol))
$\text{CH}_3\text{OH}(g)$	240.
$\text{CO}(g)$	198
$\text{H}_2(g)$	131

(i) Use the data in the table to calculate the value of the standard entropy change, ΔS° , in J/(K·mol_{rxn}), for the reaction.

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(ii) Calculate the value of ΔG° , in $\text{kJ/mol}_{\text{rxn}}$, for the reaction at 375 K. Assume that ΔH° and ΔS° are independent of temperature.

The following particle-level diagram shows a representative sample of the equilibrium mixture represented by the equation given.



(d) Use information from the particle diagram to calculate the partial pressure of CO at equilibrium when the total pressure of the equilibrium mixture is 12.0 atm.

(e) Write the expression for the equilibrium constant, K_p , for the reaction.



The reaction system represented by the equation is allowed to achieve equilibrium at a different temperature. The following table gives the partial pressure of each species in the equilibrium mixture.

Substance	Partial Pressure at Different Temperature
$\text{CH}_3\text{OH}(g)$	2.7 atm
$\text{CO}(g)$	4.2 atm
$\text{H}_2(g)$	8.4 atm

(f) Use the information in the table to calculate the value of the equilibrium constant, K_p , at the new temperature.

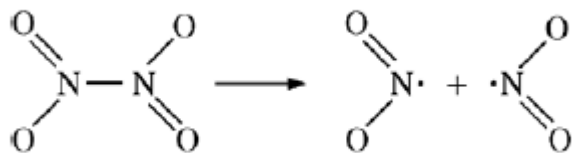
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(g) The volume of the container is rapidly doubled with no change in temperature. As equilibrium is re-established, does the number of moles of $\text{CH}_3\text{OH}(g)$ increase, decrease, or remain the same? Justify your answer by comparing the value of the reaction quotient, Q , with the value of the equilibrium constant, K_p .

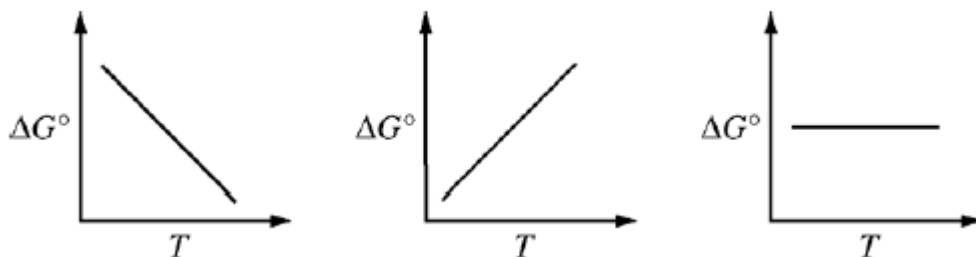
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18. Use principles of thermodynamics to answer the following questions.

- a. The gas N_2O_4 decomposes to form the gas NO_2 according to the equation below.



- Predict the sign of ΔH° for the reaction. Justify your answer.
 - Predict the sign of ΔS° for the reaction. Justify your answer.
- b. One of the diagrams below best represents the relationship between ΔG° and temperature for the reaction given in part (a). Assume that ΔH° and ΔS° are independent of temperature.



Draw a circle around the correct graph. Explain why you chose that graph in terms of the relationship $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$.

- c. A reaction mixture of N_2O_4 and NO_2 is at equilibrium. Heat is added to the mixture while the mixture is maintained at constant pressure.
- Explain why the concentration of N_2O_4 decreases.
 - The value of K_{eq} at 25°C is 5.0×10^{-3} . Will the value of K_{eq} at 100°C be greater than, less than, or equal to this value?
- d. Using the value of K_{eq} at 25°C given in part (c)(ii), predict whether the value of ΔH° is expected to be greater than, less than, or equal to the value of $T\Delta S^\circ$. Explain.

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19. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{NO}(g)$ and $\text{O}_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.



(a) If 3.2 g of $\text{O}_2(g)$ is consumed in the reaction with excess $\text{NO}(g)$, how many moles of $\text{NO}_2(g)$ are produced?

(b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{NO} (atm)	P_{O_2} (atm)	P_{NO_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.80	0.90	0.20
t_2	0.69	0.84	0.31
t_3	0.62	0.81	0.38
t_4	0.58	0.79	0.42
t_5	0.55	0.77	0.45
t_6	0.52	0.76	0.48
t_7	0.50	0.75	0.50
t_8	0.50	0.75	0.50
t_9	0.50	0.75	0.50

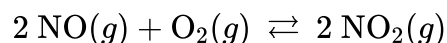
(c) At time t_3 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.

(d) At equilibrium, the container holds the most molecules of which gas, NO , O_2 , or NO_2 ? Explain your answer.

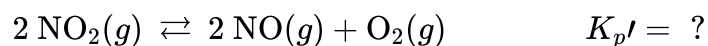
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(e) A student hypothesizes that if the temperature of the container is increased after time t_9 , the mole fraction of O_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.



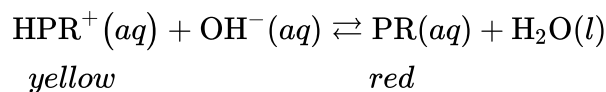
(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{NO}_2(g)$ or $\text{O}_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

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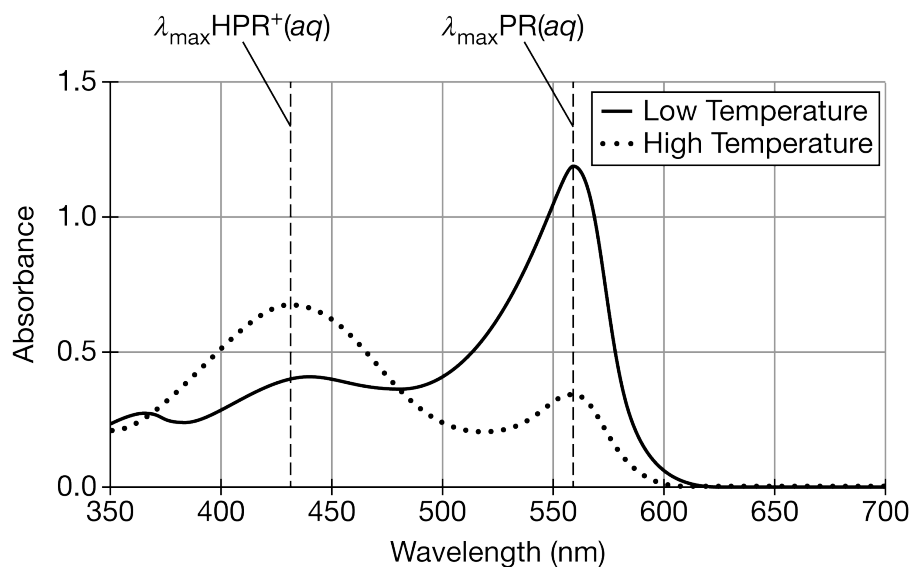
20. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The acidified form of phenol red, $\text{HPR}^+(aq)$, is yellow with a strong absorption peak at a wavelength of 435 nm. It reacts with hydroxide ions according to the equation above, producing water and phenol red (PR), which has a strong absorption peak at a wavelength of 555 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

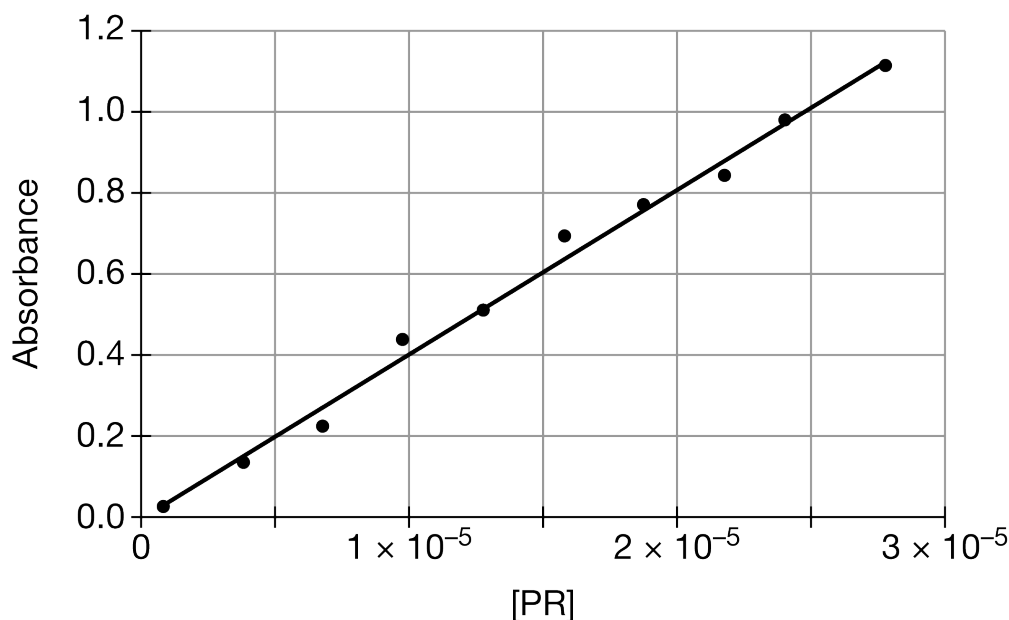
The absorption spectra of an equilibrium mixture of $\text{PR}(aq)$ and $\text{HPR}^+(aq)$, collected at two different temperatures, are shown in the following diagram.



- (b) Which peak, the one at 365 nm or the one at 435 nm, results from the absorption of photons with the greater energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at 555 nm versus the concentration of $\text{PR}(aq)$ is shown in the following graph.

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(d) Based on the best-fit line in the graph, what is the approximate absorbance value for a solution in which $[PR] = 1.5 \times 10^{-5} M$?

(e) A student measures the absorbance of a solution of PR of unknown concentration in order to determine the $[PR]$ in the solution using the calibration plot above. The cuvette is rinsed with water, then with a small amount of the solution, which is discarded. Finally, the cuvette is filled with the mixture for analysis. Explain why rinsing the cuvette with a small amount of solution is important to ensure an accurate absorbance measurement.

Over time, PR converts to a colorless molecule in the presence of OCl^- ions. To study the rate of the reaction, the student recorded the absorbance of the PR solution at 555 nm in a 2.00 M KOC1 solution. The temperature and $[OCl^-]$ remain constant throughout the reaction. The results are summarized in the table below.

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Time (hr)	Absorbance
0	0.40
0.40	0.32
0.80	0.25
1.20	0.20
1.60	0.16
2.00	0.13
2.40	0.10

(f) Determine the half-life of the reaction. Include units in your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to PR.

A second trial is conducted at the same constant temperature, but with $[\text{OCl}^-]$ changed to 0.500 M . The observed half-life of the reaction is longer than for the first trial when $[\text{OCl}^-]$ was 2.00 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction increases when $[\text{OCl}^-]$ is changed from 2.00 M to 0.500 M .

(i) Is the activation energy for the reaction when $[\text{OCl}^-]$ is 2.00 M greater than, the same as, or less than the activation energy when $[\text{OCl}^-]$ is 0.500 M ? Justify your answer.

In order to prepare the original 2.00 M solution of $\text{KOC}\text{l}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 90.6 g of $\text{KOC}\text{l}(s)$.

Step 2: Add 1.00 L of distilled water to a 2.000 L volumetric flask.

Step 3: Add the $\text{KOC}\text{l}(s)$ to the flask and swirl to dissolve the $\text{KOC}\text{l}(s)$.

Step 4: Add enough distilled water to bring the mixture to the line on the volumetric flask, inverting several times to mix.

(j) The chemical quantities used in the procedure do NOT produce $2.00\text{ M KOC}\text{l}(aq)$. Describe specifically how they should be changed in order to produce $2.00\text{ M KOC}\text{l}(aq)$.

(k) Another student is preparing a $2.00\text{ M KOC}\text{l}$ solution using the correct chemical quantities but notices that some solid remains at the bottom of the volumetric flask. Describe an action that the student should take to ensure that the remaining solid dissolves.

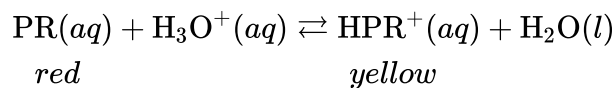
Chapter 10 Part 2: Le Chatelier's Principle FRQ

(l) When making the solution above, the student notices that the temperature increases when dissolving $\text{KCl}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{KCl}(s)$ in $1000. \text{ g}$ each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below

Solvent	X	Y
Specific Heat Capacity ($\text{J}/(\text{g} \cdot ^\circ\text{C})$)	1.8	2.2

Chapter 10 Part 2: Le Chatelier's Principle FRQ

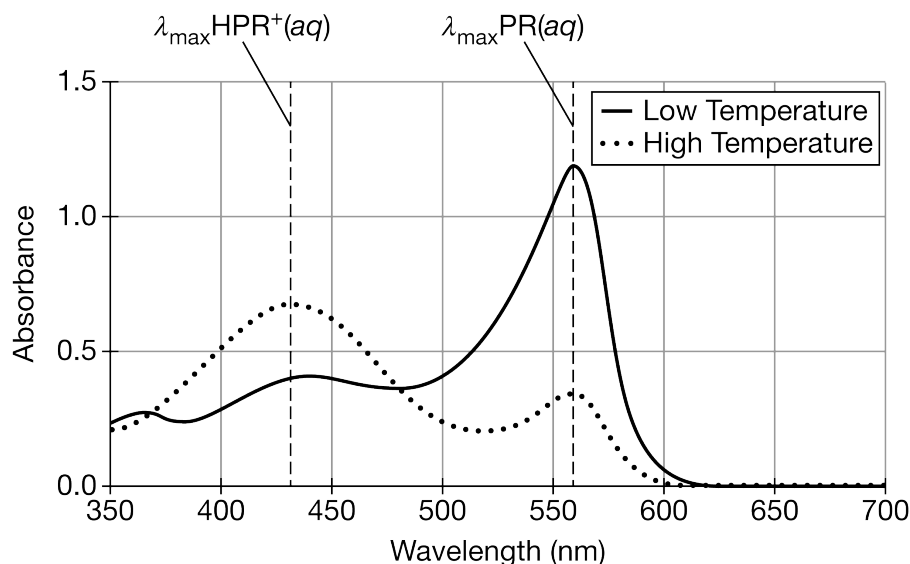
21. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Phenol red, $\text{PR}(aq)$, is a red compound that has a strong absorption peak at a wavelength of 555 nm. It reacts with hydronium ions according to the equation above, producing water and the acidified form of phenol red, $\text{HPR}^+(aq)$, which is yellow and has a strong absorption peak at a wavelength of 435 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

The absorption spectra of an equilibrium mixture of $\text{PR}(aq)$ and $\text{HPR}^+(aq)$, collected at two different temperatures, are shown in the following diagram.

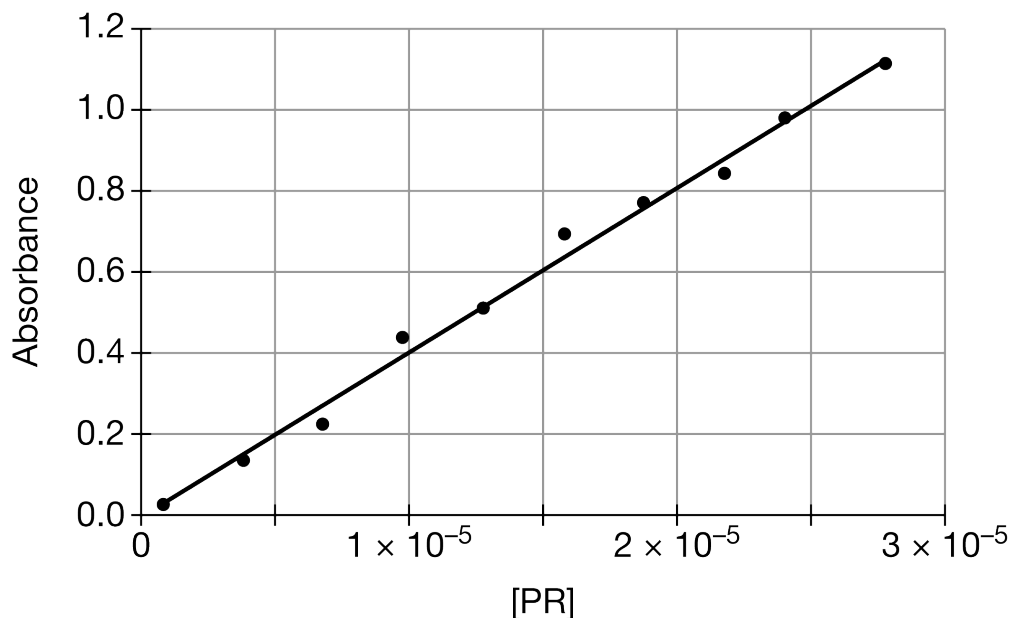


(b) Which peak, the one at 435 nm or the one at 555 nm, results from the absorption of photons with the greater energy? Justify your answer.

(c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 555 nm versus the concentration of $\text{PR}(aq)$ is shown in the following graph.

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(d) Based on the best-fit line in the graph, what is the approximate absorbance value for a solution in which $[\text{PR}] = 2.0 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a solution of PR of unknown concentration in order to determine the $[\text{PR}]$ in the solution using the calibration plot above. The outside of the cuvette is wiped clean before placing it in the spectrophotometer. Explain why this step is important to ensure an accurate absorbance measurement.

Over time, PR converts to a colorless molecule in the presence of OCl^- ions. To study the rate of the reaction, the student recorded the absorbance of the PR solution at 555 nm in a 0.50 M KOCℓ solution. The temperature and $[\text{OCl}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

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Time (min)	Absorbance
0	0.80
6	0.63
12	0.50
18	0.40
24	0.32
30	0.25
36	0.20

(f) Determine the half-life of the reaction. Include units in your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to PR.

A second trial is conducted at the same constant temperature, but with $[\text{OCl}^-]$ changed to 1.00 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OCl}^-]$ was 0.50 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OCl}^-]$ is changed from 0.50 M to 1.00 M .

(i) When $[\text{OCl}^-]$ is changed from 0.50 M to 1.00 M , does the magnitude of $\Delta H^\circ_{\text{rxn}}$, the enthalpy change of the reaction, increase, decrease, or stay the same? Justify your answer.

In order to prepare the $1.00\text{ M KOCl}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 45.3 g of $\text{KOCl}(s)$.

Step 2: Add the $\text{KOCl}(s)$ to a 1000.0 mL volumetric flask that is half-filled with distilled water and swirl to dissolve.

Step 3: Add enough distilled water to bring the volume up to the line on the flask.

(j) The chemical quantities used in the procedure do NOT produce $1.00\text{ M KOCl}(aq)$. Describe specifically how they should be changed in order to produce $1.00\text{ M KOCl}(aq)$.

(k) Another student is preparing a KOCl solution using the correct chemical quantities but notices that some undissolved solid remains at the bottom of the volumetric flask. Under these circumstances, is the concentration of $\text{KOCl}(aq)$ in the solution in the flask greater than, less than, or equal to 1.00 M ? Explain your answer.

(l) When making the solution above, the student notices that the temperature increases when dissolving $\text{KOCl}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{KOCl}(s)$ in $1000.\text{ g}$ each of two different

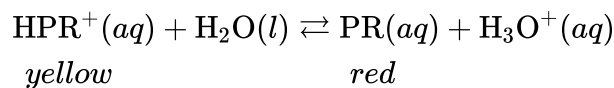
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solvents, water and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in water be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of water and solvent Y are given in the table below.

Solvent	Water	Y
Specific Heat Capacity (J/(g · °C))	4.2	2.2

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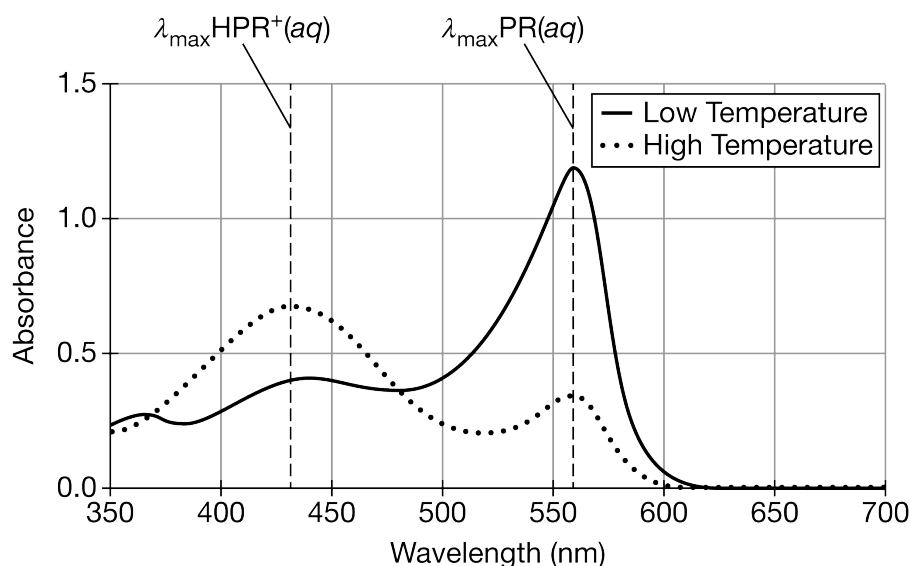
22. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The acidified form of phenol red, $\text{HPR}^+(aq)$, is yellow with a strong absorption peak at a wavelength of 435 nm. It reacts with water according to the equation above, producing hydronium ions and phenol red (PR), which has a strong absorption peak at a wavelength of 555 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

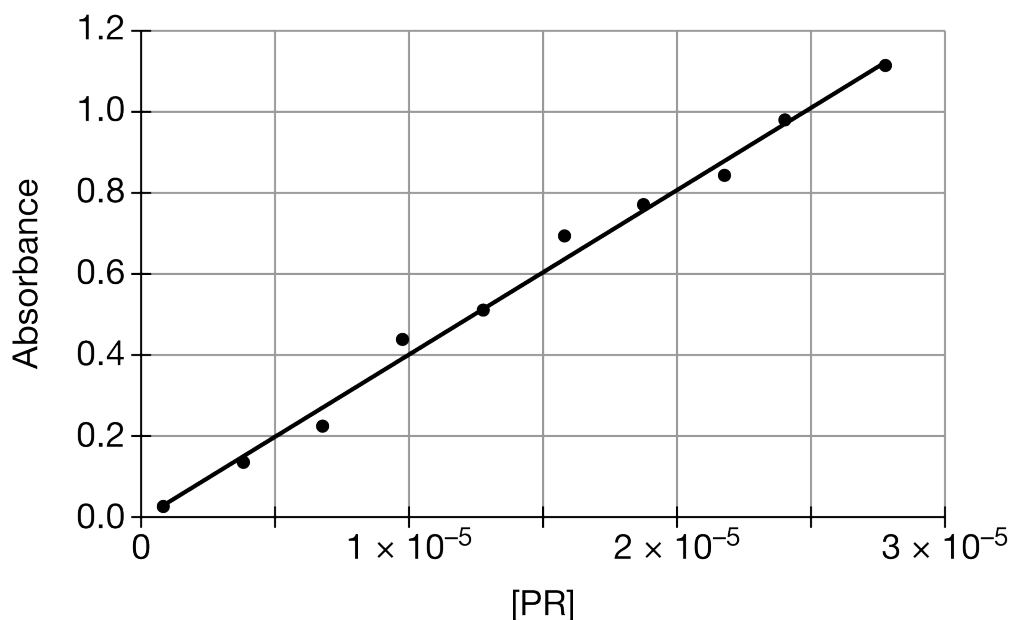
The absorption spectra of an equilibrium mixture of $\text{PR}(aq)$ and $\text{HPR}^+(aq)$, collected at two different temperatures, are shown in the following diagram.



- (b) Which peak, the one at 435 nm or the one at 555 nm, results from the absorption of photons with the lower energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at 555 nm versus the concentration of $\text{PR}(aq)$ is shown in the following graph.

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(d) Based on the best-fit line in the graph, what is the approximate absorbance value for a solution in which $[PR] = 0.5 \times 10^{-5} M$?

(e) A student measures the absorbance of a solution of PR of unknown concentration in order to determine the $[PR]$ in the solution using the calibration plot above. The cuvette used for making this measurement has the same path length as the cuvette used in making the calibration plot. Explain why this step is important to ensure an accurate absorbance measurement.

Over time, PR converts to a colorless molecule in the presence of OCl^- ions. To study the rate of the reaction, the student recorded the absorbance of the PR solution at 555 nm in a 1.00 M KOC1 solution. The temperature and $[OCl^-]$ remain constant throughout the reaction. The results are summarized in the table below.

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Time (s)	Absorbance
0	0.80
120	0.63
240	0.50
360	0.40
480	0.32
600	0.25
720	0.20

(f) Determine the half-life of the reaction. Include units in your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to PR.

A second trial is conducted at the same constant temperature, but with $[\text{OCl}^-]$ changed to 2.00 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OCl}^-]$ was 1.00 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OCl}^-]$ is changed from 1.00 M to 2.00 M .

(i) When $[\text{OCl}^-]$ is changed from 1.00 M to 2.00 M , does the magnitude of $\Delta H^\circ_{\text{rxn}}$, the enthalpy change of the reaction, increase, decrease, or stay the same? Justify your answer.

In order to prepare the 2.00 M solution of $\text{KOC}\text{l}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 18.11 g of $\text{KOC}\text{l}(s)$.

Step 2: Add the $\text{KOC}\text{l}(s)$ to a clean, dry 100.0 mL volumetric flask.

Step 3: Add about 50 mL of distilled water to the flask and swirl to dissolve the $\text{KOC}\text{l}(s)$.

Step 4: Add enough distilled water to bring the mixture to the line on the volumetric flask, inverting several times to mix.

(j) Will the chemical quantities in the procedure produce a solution that is $2.00\text{ M KOC}\text{l}(aq)$? Explain your reasoning.

(k) Before adding the $\text{KOC}\text{l}(s)$ to the volumetric flask, the student first rinsed the volumetric flask with distilled water, leaving some droplets of distilled water on the inside of the flask. As a result of this step, will the concentration of $\text{KOC}\text{l}(aq)$ be greater than, less than, or equal to 2.00 M ? Explain your answer.

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(l) When making the solution above, the student notices that the temperature increases when dissolving $\text{KCl}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{KCl}(s)$ in $1000. \text{ g}$ each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below.

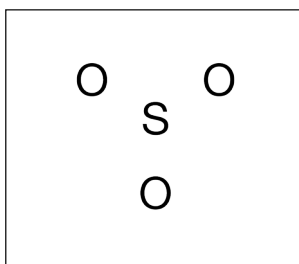
Solvent	X	Y
Specific Heat Capacity ($\text{J}/(\text{g} \cdot ^\circ\text{C})$)	2.2	1.8

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23. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

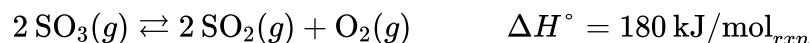
Sulfur trioxide, SO_3 , is commercially produced on a massive scale for use in manufacturing. It is also produced as a by-product of burning fossil fuels and is an air pollutant involved in the formation of acid rain.

(a) In the box below, complete one of the valid Lewis electron-dot diagrams for the SO_3 molecule. Be sure to include all valence electrons, including lone pairs.



(b) Predict the shape of the SO_3 molecule based on the diagram you completed above.

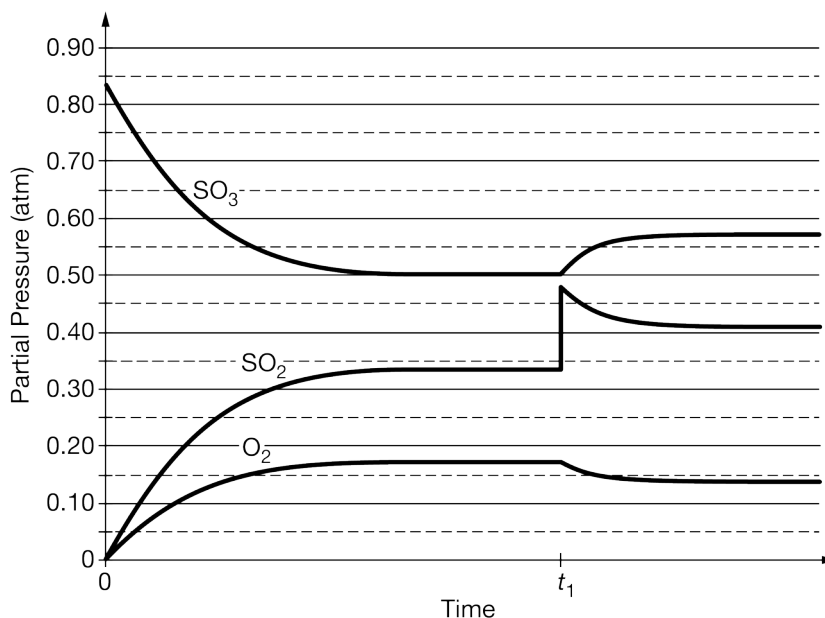
Under some conditions, $\text{SO}_3(g)$ decomposes to produce $\text{SO}_2(g)$ and $\text{O}_2(g)$, as represented by the equation below.



(c) Under what temperature conditions will the reaction be thermodynamically favored: high temperatures, low temperatures, all temperatures, or no temperatures? Justify your answer in terms of the change in enthalpy and the change in entropy during the reaction.

A chemist fills a rigid vessel with $\text{SO}_3(g)$ at a certain temperature until the pressure of $\text{SO}_3(g)$ in the container is 0.83 atm. The $\text{SO}_3(g)$ decomposes as the partial pressures of $\text{SO}_3(g)$, $\text{SO}_2(g)$, and $\text{O}_2(g)$ in the container are monitored over time, as shown in the following graph. A change is made to the system at time t_1 .

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(d) Answer the following based on the graph.

(i) Draw an X on the time axis to show when the system first reached equilibrium.

(ii) Describe the change that was made to the system at time t_1 .

(iii) After the change was made at time t_1 , the partial pressure of $\text{SO}_3(g)$ increased while the partial pressure of $\text{O}_2(g)$ decreased. Explain this observation.

(e) Use the information above to answer the following.

(i) Write the expression for the equilibrium constant, K_p , for the reaction.

(ii) Determine the value of K_p for the reaction.

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24. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

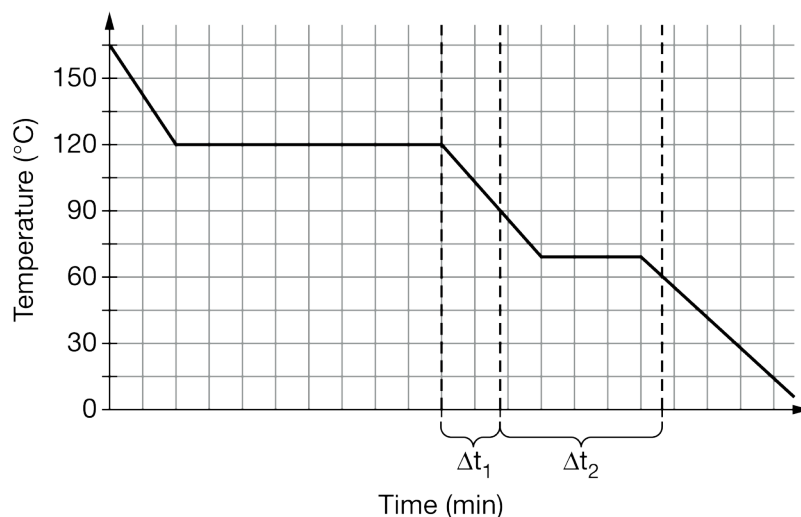
Answer the following questions about the compounds and reactions of Vb, a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VbO_2 , an oxide of Vb.



- (a) Identify the hybridization of the valence orbitals of the Vb atom in VbO_2 .

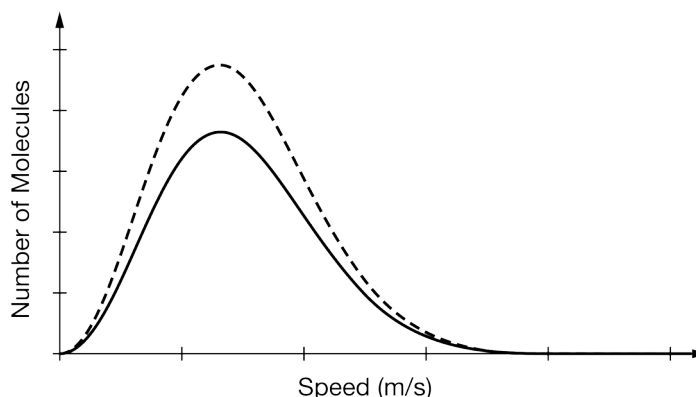
A student investigates the physical properties of VbO_2 and finds the following graph where thermal energy is removed at a constant rate from a sample of VbO_2 , which is initially a gas at 165°C . The pressure is held constant at 1.00 atm throughout the cooling process.



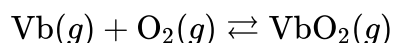
- (b) The student determines that the specific heat of $\text{VbO}_2(l)$ is $2.0 \text{ J}/(\text{g} \cdot ^\circ\text{C})$. Determine the amount of thermal energy, in J, that must be removed from a 35 g sample of $\text{VbO}_2(l)$ to reduce the temperature of the sample from $110.^\circ\text{C}$ to $80.^\circ\text{C}$.
- (c) The two intervals marked on the graph correspond to the time required for a temperature decrease of $30.^\circ\text{C}$. Given that the sample is cooled at a constant rate, explain why $\Delta t_2 > \Delta t_1$.
- (d) The student claims that the intermolecular attractions in liquid SO_2 ($\Delta H_{\text{vap}}^\circ = 25 \text{ kJ/mol}$) are stronger than those in liquid VbO_2 ($\Delta H_{\text{vap}}^\circ = 32 \text{ kJ/mol}$). Do you agree or disagree with the student's claim? Justify your answer in terms of the relative values of $\Delta H_{\text{vap}}^\circ$.

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(e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VbO_2 particles at 165°C . The student incorrectly sketches the dashed line to show the distribution of VbO_2 particles when the temperature decreases to 135°C . Explain the student's error.



The student injects $\text{Vb}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vb}(g)$ is 6.5 atm , and the initial pressure of $\text{O}_2(g)$ is 4.0 atm . $\text{Vb}(g)$ and $\text{O}_2(g)$ react to produce $\text{VbO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 1.5 atm of $\text{VbO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vb}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a larger volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{Vb}(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vb}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vb}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	8.0×10^{-4}
2	0.20	0.30	1.2×10^{-3}
3	0.10	0.30	3.0×10^{-4}

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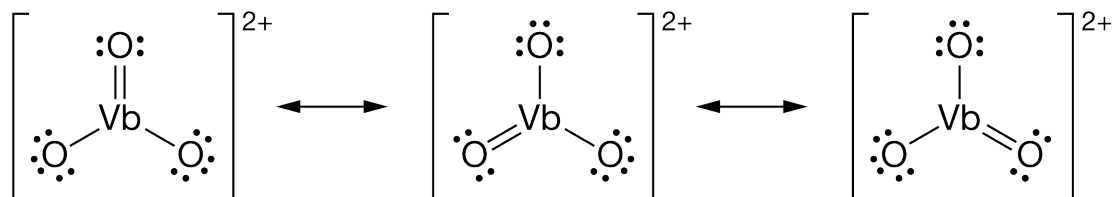
- (i) Based on the data, determine the order of the reaction with respect to $\text{O}_2(g)$. Justify your answer.
- (j) The reaction is known to be first order with respect to $\text{Vb}(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



- (k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element Vb forms ions with formula VbO_3^{2+} . The following Lewis diagrams represent the VbO_3^{2+} ion.



- (l) A student claims that VbO_3^{2+} contains two $\text{Vb} - \text{O}$ bonds of equal strength and one $\text{Vb} - \text{O}$ bond that is stronger than the other two. Do you agree or disagree? Justify your answer.

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25. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

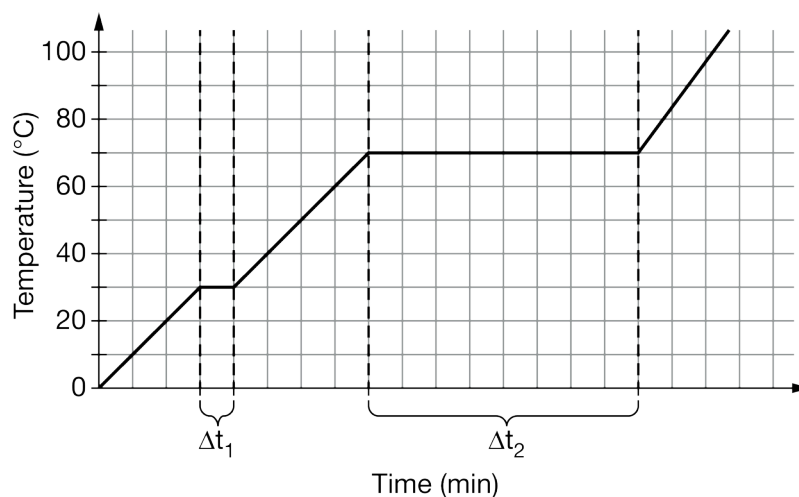
Answer the following questions about the compounds and reactions of Vb, a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VbO₂, an oxide of Vb.



- (a) Identify the hybridization of the valence orbitals of the Vb atom in VbO₂.

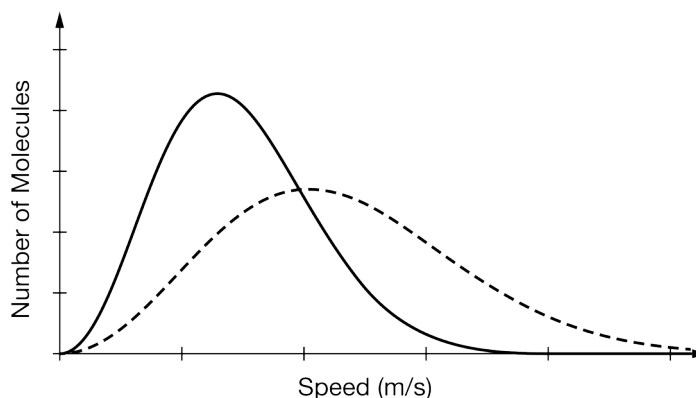
A student investigates the physical properties of VbO₂ and finds the following graph where thermal energy is added at a constant rate to a sample of VbO₂, which is initially a solid at 0 °C. The pressure is held constant at 1.00 atm throughout the heating process.



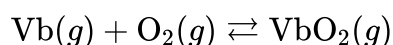
- (b) The student determines that the specific heat of VbO₂(l) is 2.0 J/(g · °C). Determine the amount of thermal energy, in J, that must be added to a 5.5 g sample of VbO₂(l) to raise the temperature from 40. °C to 60. °C.
- (c) Two intervals are marked on the graph where the temperature of the sample remains constant. Given that the sample is heated at a constant rate, explain why $\Delta t_2 > \Delta t_1$.
- (d) The student claims that the intermolecular attractions in liquid H₂O (boiling point = 100 °C) are weaker than those in liquid VbO₂. Do you agree or disagree with the student's claim? Justify your answer in terms of the relative boiling points.
- (e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VbO₂ at 110 °C.

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The student incorrectly sketches the dashed line to show the distribution of VbO_2 particles when the temperature decreases to 80°C . Explain the student's error.



The student injects $\text{Vb}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vb}(g)$ is 7.0 atm , and the initial pressure of $\text{O}_2(g)$ is 5.5 atm . $\text{Vb}(g)$ and $\text{O}_2(g)$ react to produce $\text{VbO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 2.5 atm of $\text{VbO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vb}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a smaller volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{VbO}_2(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

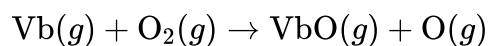
To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vb}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vb}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	8.0×10^{-4}
2	0.20	0.30	1.2×10^{-3}
3	0.10	0.30	1.2×10^{-3}

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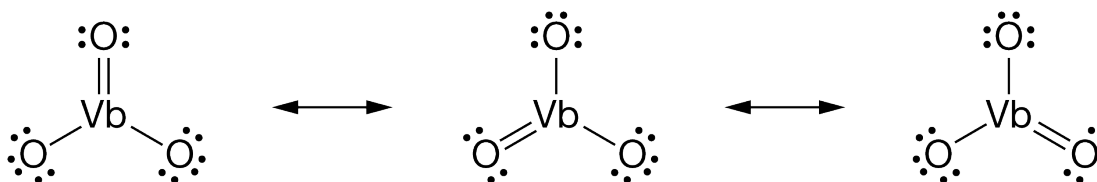
- (i) Based on the data, determine the order of the reaction with respect to $\text{O}_2(g)$. Justify your answer.
- (j) The reaction is known to be first order with respect to $\text{Vb}(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



- (k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element Vb forms a molecule with formula VbO_3 . The following Lewis diagrams represent the VbO_3 molecule.



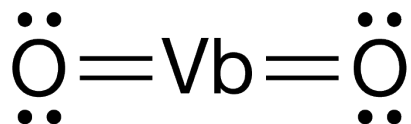
- (l) A student claims that VbO_3 contains all three $\text{Vb} - \text{O}$ bonds of equal length. Do you agree or disagree? Justify your answer.

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26. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

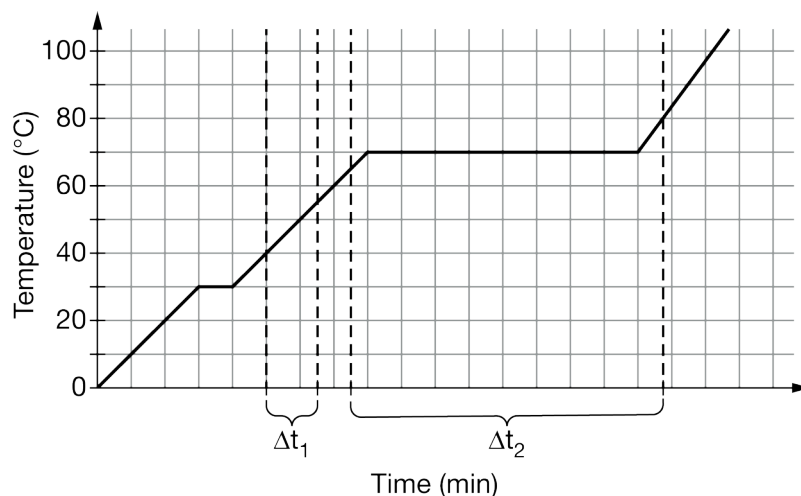
Answer the following questions about the compounds and reactions of Vb, a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VbO₂, an oxide of Vb.



- (a) Identify the hybridization of the valence orbitals of the Vb atom in VbO₂.

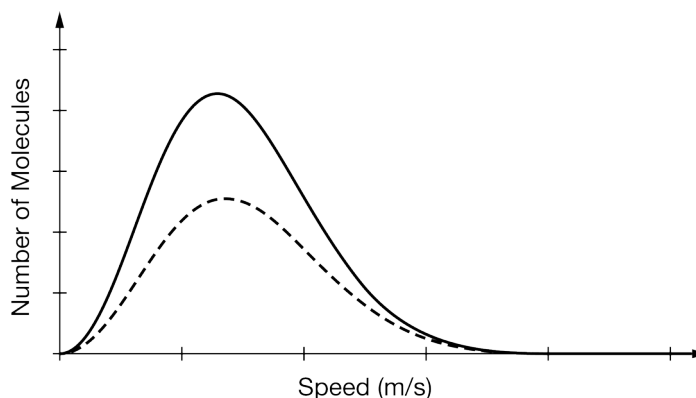
A student investigates the physical properties of VbO₂ and finds the following graph where thermal energy is added at a constant rate to a sample of VbO₂, which is initially a solid at 0 °C. The pressure is held constant at 1.00 atm throughout the heating process.



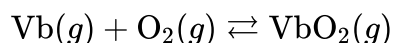
- (b) The student determines that the specific heat of VbO₂ is 2.0 J/(g · °C). Determine the amount of thermal energy, in J, that must be added to a 4.5 g sample of VbO₂(l) to raise the temperature from 40. °C to 55 °C.
- (c) The two intervals marked on the graph correspond to the time required for the temperature of the sample to increase by 15 °C. Given that the sample is heated at a constant rate, explain why $\Delta t_2 > \Delta t_1$.
- (d) The student claims that the intermolecular attractions in liquid H₂O ($\Delta H_{vap}^\circ = 40$ kJ/mol) are stronger than those in liquid VbO₂ ($\Delta H_{vap}^\circ = 32$ kJ/mol). Do you agree or disagree with the student's claim? Justify your answer in terms of the relative values of ΔH_{vap}° .
- (e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VbO₂ at 110 °C.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The student incorrectly sketches the dashed line to show the distribution of VbO_2 particles when the temperature decreases to 80°C . Explain the student's error.



The student injects $\text{Vb}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vb}(g)$ is 6.0 atm , and the initial pressure of $\text{O}_2(g)$ is 4.0 atm . $\text{Vb}(g)$ and $\text{O}_2(g)$ react to produce $\text{VbO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 2.5 atm of $\text{VbO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vb}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a larger volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{VbO}_2(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vb}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

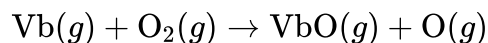
Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vb}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	8.0×10^{-4}
2	0.20	0.30	1.2×10^{-3}
3	0.10	0.30	6.0×10^{-4}

(i) Based on the data, determine the order of the reaction with respect to $\text{O}_2(g)$. Justify your answer.

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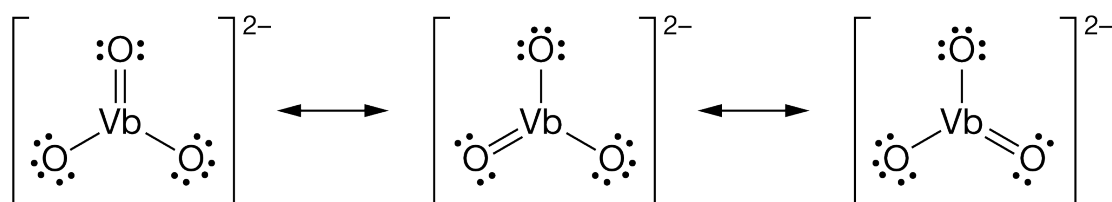
(j) The reaction is known to be first order with respect to $\text{Vb}(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



(k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element Vb forms ions with formula VbO_3^{2-} . The following Lewis diagrams represent the VbO_3^{2-} ion.



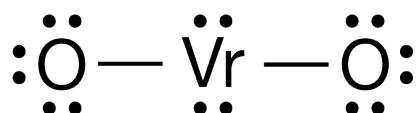
(l) A student claims that VbO_3^{2-} contains two $\text{Vb} - \text{O}$ bonds of equal length and one $\text{Vb} - \text{O}$ bond that is shorter than the other two. Do you agree or disagree? Justify your answer.

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27. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

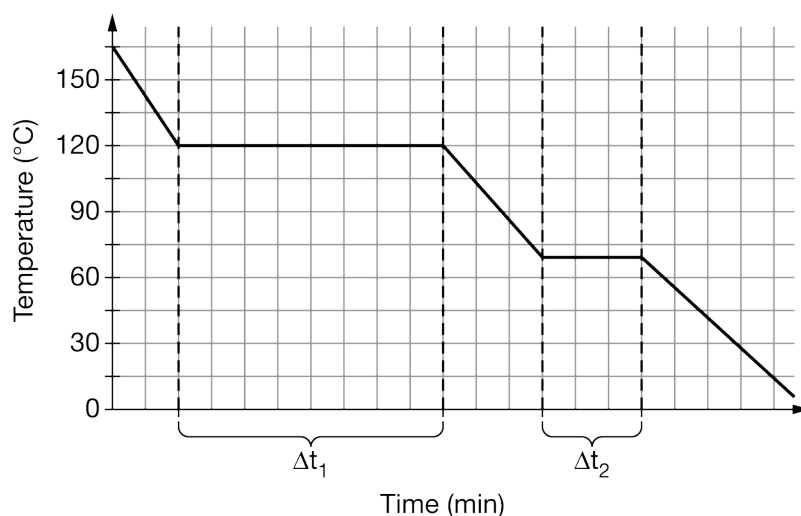
Answer the following questions about the compounds and reactions of Vr , a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VrO_2 , an oxide of Vr .



- (a) Based on VSEPR theory, predict the approximate bond angle in VrO_2 .

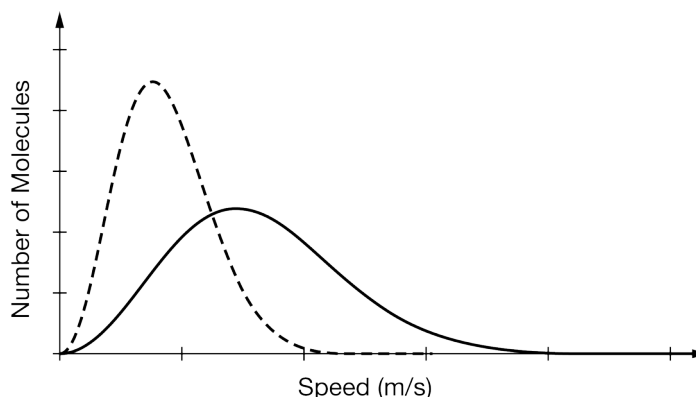
A student investigates the physical properties of VrO_2 and finds the following graph where thermal energy is removed at a constant rate from a sample of VrO_2 , which is initially a gas at 165°C . The pressure is held constant at 1.00 atm throughout the cooling process.



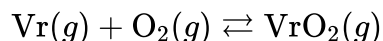
- (b) The student determines that the specific heat of $\text{VrO}_2(l)$ is $2.0\text{ J}/(\text{g} \cdot ^\circ\text{C})$. Determine the amount of thermal energy, in J, that must be removed from a 25 g sample of $\text{VrO}_2(l)$ to reduce the temperature from 115°C to 105°C .
- (c) Two intervals are marked on the graph where the temperature of the sample remains constant. Given that the sample is cooled at a constant rate, explain why $\Delta t_1 > \Delta t_2$.
- (d) The student claims that the intermolecular attractions in liquid SO_2 (boiling point = -10°C) are weaker than those in liquid VrO_2 . Do you agree or disagree with the student's claim? Justify your answer in terms of the relative boiling points.

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(e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VrO_2 at 165°C . The student incorrectly sketches the dashed line to show the distribution of VrO_2 particles when the temperature increases to 205°C . Explain the student's error.



The student injects $\text{Vr}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vr}(g)$ is 7.0 atm , and the initial pressure of $\text{O}_2(g)$ is 5.5 atm . $\text{Vr}(g)$ and $\text{O}_2(g)$ react to produce $\text{VrO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 1.5 atm of $\text{VrO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vr}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a smaller volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{Vr}(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

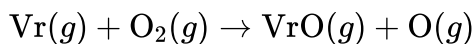
To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vr}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vr}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	1.2×10^{-3}
2	0.20	0.40	1.2×10^{-3}
3	0.10	0.40	6.0×10^{-4}

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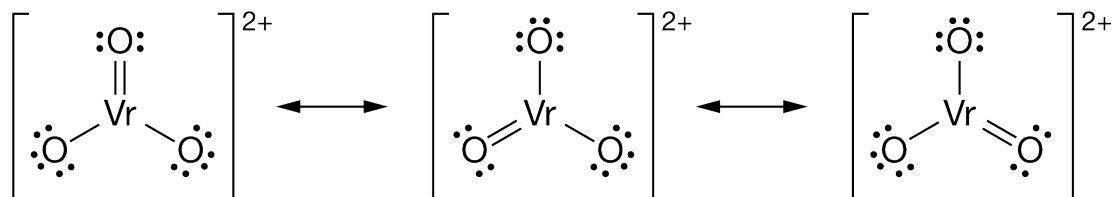
- (i) Based on the data, determine the order of the reaction with respect to $\text{Vr}(g)$. Justify your answer.
- (j) The reaction was found to be first order with respect to $\text{O}_2(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



- (k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element Vr forms ions with formula VrO_3^{2+} . The following Lewis diagrams represent the VrO_3^{2+} ion.



- (l) A student claims that VrO_3^{2+} contains three Vr — O bonds of equal strength. Do you agree or disagree? Justify your answer.